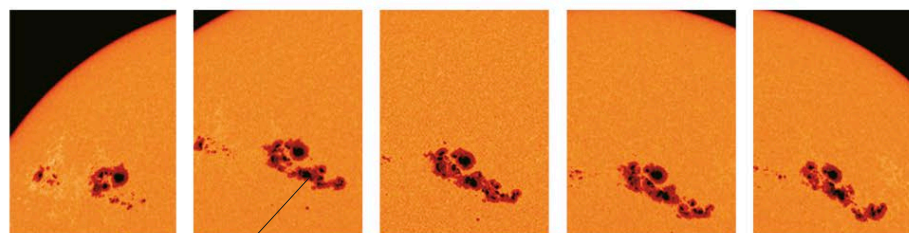


ROTATION AND MAGNETISM

The pressures and temperatures within stars mean they are composed of plasma (see p.30). Within this ionized matter, negatively charged electrons travel free from the positively charged ions. This has a profound effect on magnetic fields, since charged particles do not cross magnetic field lines easily. Magnetic field lines can dictate the movement of stellar material, but the movement of the plasma can also affect magnetic fields.

All stars rotate, and some spin so fast they bulge out at the equator and are very flattened at the poles. As stars rotate, magnetic field lines are carried around by the plasma. This “winds up” the magnetic field and creates pockets of intense magnetic flux where field lines are brought close together. The movement of stellar material and the transfer of heat is restricted in these areas, so they are appreciably cooler than the surrounding material. As they are cooler, they appear dark against the rest of the photosphere. Dark star spots on the surface of stars are areas of intense activity, because the buildup of heat around them can suddenly be released as flares.

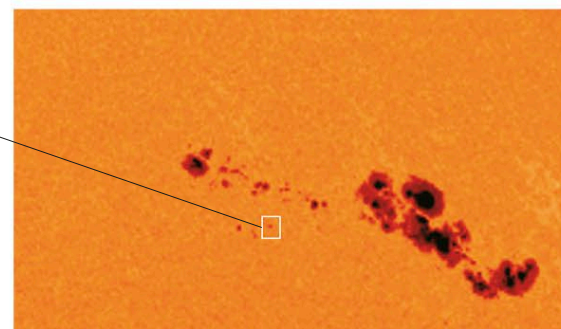


sun-spot group has rotated from previous position

region of equal area to the Earth

SOLAR ROTATION

As the Sun rotates, sun-spot groups are observed traveling across its disk. Main-sequence stars rotate differentially, with material at the equator rotating faster than that at the poles. On the Sun, sunspots closer to the equator travel across the solar disk more rapidly.



ARTHUR EDDINGTON



British astronomer Arthur Stanley Eddington (1882–1944) studied the internal structure of stars and derived a mass–luminosity relationship for main-sequence stars. In 1926, he published *The Internal Constitution of Stars*, in which he suggested that nuclear reactions were the power source of stars. While working at the Royal Greenwich Observatory, Eddington led two expeditions to view total solar eclipses and in 1919 provided evidence for the theory of general relativity. Eddington also calculated the abundance of hydrogen within stars and developed a model for Cepheid variable pulsation (see p.282). He became Plumian professor of astronomy at Cambridge in 1913 and director of the Cambridge Observatory in 1914. He was knighted in 1930.

THE MAIN SEQUENCE

A star enters the main sequence when it starts to burn hydrogen in its core. As soon as the nuclear reactions instigating this process begin, it is said to be at age zero on the main sequence. A star's life on the main sequence is very stable, with the pressure from the nuclear reactions in its core being balanced by its gravity trying to compress all of its mass into the center. A star will spend most of its life on the main sequence, and consequently about 90 percent of the stars observed in the sky are main-sequence stars. A star's time on the main sequence is dependent on its mass. The more massive the star, the hotter and denser its core and the faster it will convert hydrogen into helium. The Sun is a relatively small main-sequence star and will be on the main sequence for about 10 billion years. A 10-solar-mass star will be on the main sequence for only 10 million years. While on the main sequence, a star will conform to the mass–luminosity relation, which means that the absolute magnitude or luminosity of a star will give an indication of its mass. As it converts hydrogen into helium, a star's chemical composition and internal structure will change and it will move slightly to the right of its zero-age position on the H–R diagram (below). As soon as the hydrogen in the core is depleted and hydrogen burning in the atmosphere begins, the star leaves the main sequence (see p.236).

DIAGONAL PATH

The main sequence is a diagonal curving path of stars on the Hertzsprung–Russell diagram, a simplified version of which is shown here (see also p.232). The curve runs from bottom right (low mass and cool) to top left (massive and hot). Each star has a “zero-age” position (a point on the curve indicating its mass and temperature). It hardly strays at all from this position during its time on the main sequence.

